

Finite Fields

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Problem 1. Let F be a finite field. Let $\mathbb{Z} \rightarrow F$ be the unique ring homomorphism from the integers. Explain why the kernel is (p) for some prime p . Conclude that F contains $\mathbb{F}_p$. Explain why F is finite dimensional as an $\mathbb{F}_p$ vector space. Explain why $|F| = p^n$ for some n .

Proof. The kernel is nonzero, because if it was zero that would mean $\mathbb{Z} \subset F \implies |F| = |\mathbb{Z}|$ and thus not finite.

Since F is a field and thus a integral domain, that means that any subring of F is also a integral domain and thus the kernel of the map is a prime ideal (p) . Thus the map factors through $\mathbb{Z} \rightarrow \frac{\mathbb{Z}}{(p)} = \mathbb{F}_p$ and the map $\mathbb{F}_p \rightarrow F$ is a inclusion. Thus $\mathbb{F}_p \subseteq F$.

$$\begin{array}{ccc} \mathbb{Z} & \twoheadrightarrow & \mathbb{Z}/(p) \cong \mathbb{F}_p \\ & \searrow & \downarrow \\ & & F \end{array}$$

Since $\mathbb{F}_p \subseteq F$, F can be represented as a vectorspace over $\mathbb{F}_p$.

Assume the contrary, that F is not finite dimensional over $\mathbb{F}_p$, then there are an infinite amount of linearly independent vectors $f \in F$ over $\mathbb{F}_p$, which contradicts the finiteness of F .

Thus as a vectorspace, $F \cong \mathbb{F}_p^n$. Thus the number of elements in F can be calculated as the number of possible entries in each component of the vector, ie p entries for each of the n components. Thus $|F| = p^n$ $\square$

Problem 2. F a finite field of order $q := p^n$. We showed on the last homework that F^* is cyclic. Show that $f(X) := X^q - X$ has distinct roots in any field containing F . Explain why its roots are exactly the elements of F .

Proof. Assume the contrary, $f(X)$ has repeated roots over any field extension of $\mathbb{F}_p$. Let r be a generic repeated root of f . By the definition of a repeated root, $f(r) = 0$ and $f'(r) = 0$. Thus $r^q - r = 0 \implies r(r^{q-1} - 1) = 0 \implies r = 0$ or $r^{q-1} = 1$ and $qr^{q-1} - 1 = 0 \implies r^{q-1} = \frac{1}{q}$. Unless $q = 1$, which contradicts F is a finite field with 1, this is impossible as r cannot be both 0 or a $q-1$ th root of 1 and a $q-1$ th root of $\frac{1}{q}$. Thus by contradiction, f doesn't have repeated roots over any field extension of $\mathbb{F}_p$. $\square$

Problem 3. Let $f(X)$ be a polynomial in $K[X]$, K field. By a "splitting field" for f , we mean a field extension L/K such that f has all its roots in L , and L is generated over K by these roots. Explain why any polynomial has a splitting field, and why any two splitting fields are isomorphic (its the same ideas we used to bound the size of $\text{Gal}(L/K)$ in earlier homework)

Proof. For any polynomial f . Let $r_1, \dots, r_n$ be all the roots of f . Factor f into its irreducible factors, so $f(x) = g_1(x) \dots g_k(x)$. Let $K_0 = K$. Let $K_1 = K_0[x]/(g_1)$. Now divide f by g_1 . Then $\forall i$, do $K_i = K_{i-1}[X]/(g_i)$. There should be at most $\deg f$ steps and f is a polynomial. Finally in the last step the last field should have all roots of f and thus be the splitting field of f . Thus by construction a splitting field exists.

$\forall f \in K[x]$. Let $\varphi : K \rightarrow K$ be an isomorphism. Let L and L' be the splitting fields of f and $\varphi(f)$ respectively. Let p be an irreducible factor of f . Let $\alpha_1 \in L$ be a root of p and $\beta_1 \in L'$ be a root of $\varphi(p)$. Thus $K[\alpha_1] \cong \frac{K[X]}{(p(x))}$ and $K[\beta_1] = \frac{K[X]}{(\varphi(p)(x))}$ since $p \cong \varphi(p)$, thus, $(p) \cong (\varphi(p))$ thus $K[\alpha_1] \cong \frac{K[X]}{(p(x))} \cong \frac{K[X]}{(\varphi(p)(x))} \cong K[\beta_1]$. Apply induction on the number of irreducible factors, then we will have an isomorphism between $K[\alpha_1, \dots, \alpha_k] \rightarrow K[\beta_1, \dots, \beta_k]$ that restricts to φ on K . All the roots of the irreducible factors of f are in $K[\alpha_1, \dots, \alpha_k]$ thus f splits over $K[\alpha_1, \dots, \alpha_k]$. Thus $\varphi(f)$ splits over $K[\beta_1, \dots, \beta_k]$. It follows, that $L = K[\alpha_1, \dots, \alpha_k]$ and $L' = K[\beta_1, \dots, \beta_k]$. $\square$

Problem 4. Let F be a finite field of order $q := p^n$. Explain why F is the splitting field, over F_p , of $X^q - X$. Now the uniqueness in 3) proves that F is unique (any two fields of this order are isomorphic).

Proof. $\forall f \in F_q$. If $f = 0$, f is a solution to $X^q - X$. If $f \neq 0$, $f \in F_q^* = \mu_{q-1}$. Thus $f^q - f = f - f = 0$ and thus f is a solution. Thus $\forall f \in F$, f is a solution. Moreover since all the f are distinct and there are q f s, and $x^q - x$ has degree q . $x^q - x$ has all its roots in F and since each element of F is a root and they are distinct no intermediate field between F_p and F has all the roots. Thus F is the splitting field of $x^q - x$ and thus unique. $\square$

Problem 5. Let F be a finite field of order q . Let $s : F \rightarrow F$ be the map $x \rightarrow x^p$. Explain why this is an element of $G := \text{Gal}(F/F_p)$. Explain why F_p is the fixed field of s . Explain why s has order n . Conclude that G is cyclic generated by s and that the extension is Galois.

Proof. $\forall x, y \in F$, $s(x+y) = (x+y)^p = x^p + y^p + \sum_{j=1}^{p-1} \binom{p}{j} x^j y^{p-j} = x^p + y^p$, because F has characteristic p , and $s(xy) = x^p y^p$. Thus s is a ring homomorphism (more descriptively a field homomorphism).

Let $k \in \ker s$, $s(k) = k^p = 0$. Since F is a field and thus an integral domain, it means, either $k = 0$ or $k^{p-1} = 0$. If $k = 0$ we are done. If $k^{p-1} = 0$, then since F is an integral domain, either $k = 0$ or $k^{p-2} = 0$. We can continue this chain of reasoning, till we see that $k^2 = 0$ and finally since F is an integral domain, $k = 0$. Thus s is injective. Thus s is surjective since $|F| = |F|$. Thus s is an automorphism of F .

Moreover since $F_p^* = \mu_{p-1}$ every nonzero element of F_p has order which divides $p-1$ and $0^p = 0$ since F_p is an integral domain, thus $\forall f \in F_p$, $s(f^p) = f^p = f * f^{p-1} = f$. Thus $s \in \text{Gal}(F/F_p)$.

$\forall x \in F$, $s^n(x) = x^{p^n}$. If $x = 0$, $s^n(x) = 0$, since F is an integral domain. If $x \neq 0$, $x \in F^*$ and since $F^* \cong \mu_{q-1} = \mu_{p^n-1}$, thus $x^{p^n} = x^{p^n-1}x = x$. Thus $s^n = \text{id}$.

$\forall m < n$, let $g \in F$ be the generator of F^* , $s^m(g) = g^{p^m} = g^{p^m-1}g \neq g$, because $o(g) = p^n - 1 > p^m - 1$.

Thus $|\langle s \rangle|$ is n . But since $|\text{Gal}(F/F_p)| \leq [F : F_p] = n$, $|\text{Gal}(F/F_p)| = n$ and s generates $\text{Gal}(F/F_p)$, because $\langle s \rangle \leq \text{Gal}(F/F_p)$. $\square$

Problem 6. Now we prove the existence of a finite field of order q . Let F be the splitting field, over F_p , of $X^q - X$. Let $S : F \rightarrow F$ send $x \rightarrow x^q$. Explain why this is an automorphism. Explain why the set of elements fixed by S is a subfield, and why this subfield has order q . Prove this subfield is F .

Proof. $F_p \subseteq F \implies \text{char} F = p$. $\forall x, y \in F$, $S(x+y) = (x+y)^q = x^q + y^q + \sum_{n=1}^{q-1} \binom{q}{n} x^n y^{q-n} x^q + y^q + \sum_{n=1}^{q-1} \frac{q!}{n!(q-n)!} x^n y^{q-n} = x^q + y^q + \sum_{n=1}^{q-1} q \frac{(q-1)!}{n!(q-n)!} x^n y^{q-n}$. Since we are in $\text{char}(p)$ and $q = p^n$ for some n by problem 1, $q \frac{(q-1)!}{n!(q-n)!} x^n y^{q-n} = 0$. Thus $S(x+y) = x^q + y^q$. $\forall x, y \in F$, $S(xy) = x^q y^q$. Thus S is a ring homomorphism and thus injective and thus surjective, ($|F| = |F|$), thus is a automorphism of F .

Let $F^S := \{x \in F \mid S(x) = x\}$. $\forall u, v \in F^S$, $S(u+v) = S(u) + S(v) = u + v \implies u + v \in F^S$. $S(uv) = S(u)S(v) = uv \implies uv \in F^S$ and $S(u^{-1}) = S(u)^{-1} = u^{-1} \implies u^{-1} \in F^S$. Thus F^S is a subfield.

F consists of all roots of $x^q - x$, because its the splitting field of $x^q - x$. However all roots of $x^q - x$ are also fixed points of S . Thus all roots of $x^q - x$ are in F^S . $x^q - x$ has distinct roots. Thus $|F^S| = q$. Since all the roots of $x^q - x$ are in F^S . Thus F^S is a splitting field of $x^q - x$. By the uniqueness of the splitting field, $F^S = F$. $\square$

Problem 7. This is a general result: Let $K \subset L$ be a finite extension of fields (not necessarily finite fields). Let H be a subset of $\text{Gal}(L/K)$ such that the fixed field $L^H := \{l \in L \mid h(l) = l \forall h \in H\}$ is K . For any a in L , let g be the polynomial whose distinct roots are the elements of Ha (apply each element of H to a and take the subset of L you get). Explain why g is a polynomial over K , why its irreducible over K , and why it has distinct roots. Explain (using our basic Ext homework) why L over K is "Galois", i.e. the Galois group has the same order as the degree of the extension. Conclude H is this Galois group, and has order the degree of L over K

Proof. g has distinct roots because its roots are by definition elements of the set Ha and thus distinct, by the definition of $g(x) = \prod_{\alpha \in Ha} (x - \alpha)$. $\forall h \in H$, $h(g(x))$ just reorders the product, thus $h(g(x)) = g(x)$. Thus $g(x)$ is H invariant. Thus the coefficients of g are H invariant and thus live L^H . Suppose, $g(x) = h(x)k(x)$ where $h(x), k(x) \in K[X]$. Since a is a root of $g(x)$, a must be a root of $h(x)$ or $g(x)$. WLOG, suppose a is a root of $h(x)$. By homomorphism properties, $\forall \sigma \in H$, $h(\sigma(a)) = \sigma(h(a)) = 0$, thus $\sigma(a)$ is a root of $h(x)$. Thus all roots of $g(x)$ are roots of $h(x)$. Thus $k(x)$ must be unit and $h(x) = g(x)$. Thus $g(x)$ is a irreducible over L^H .

Suppose $H \leq \text{Gal}(L/K)$ and $L^H = K$. $\forall l \in L$. Define, $p_l(x) := \prod_{\alpha \in H \cdot l} (x - \alpha)$. By the above, p_l is irreducible over $L^H = K$ and has distinct roots, more specifically distinct roots in L , since $H \leq \text{Gal}(L/K)$ and elements of H are automorphisms of L . Thus L/K is Galois. Moreover since $|H| \leq |\text{Gal}(L/K)| = [L : K] \leq \deg p_l \leq |H| \implies |H| = |\text{Gal}(L/K)| = [L : K]$ and $H = \text{Gal}(L/K)$. $\square$

Problem 8. Let F be a finite field, of order $q = p^r$. (note by the above we know F is unique). Show that F_p is the fixed field of the Frobenius $x \rightarrow x^p$. Conclude that this aut has order r , and that F over F_p is Galois, with cyclic Galois group μ_r generated by this Frobenius automorphism.

Proof. $K := \{x \in F \mid x^p = x\}$. Thus all the elements of K are solutions to $x^p - x$. Thus K is the splitting field of $x^p - x$ over F_p . By problem 6, $|K| = p$ thus $K = F_p$, since $F_p \subseteq K$. $\square$

Problem 9. Consider $X^q - X$, for $q = p^r$. Explain why this has distinct roots over F_p , and that it factors over $F_p[X]$ into the product of all irreducible polynomials over F_p whose degree divides r each poly appearing with multiplicity one.

Proof. By problem 2, $x^q - x$ has distinct roots over F_p . $x^q - x = x(x^{q-1} - 1)$. $x^q - x$ is not irreducible and thus can be broken into irreducible factors. $x^q - x = f_1(x) \dots f_k(x)$. Roots of $x^q - x$ are roots of one $f_i(x)$. Thus $F_p[X]/f_i(x) \subset F_q \implies \deg f_i | [F_q : F_p] \implies \deg f_i | r$.

Suppose g is a irreducible polynomial where $d = \deg g$ which divides r . Then $F = F_p[X]/(g) = F_p[\alpha]$. $F^* = \mu_{p^d-1}$. Thus $\alpha^{p^d-1} = 1 \implies \alpha^{p^d} = \alpha$. Since $d | n$ $\alpha^{p^n} = \alpha^{p^{d(n/d)}} = \alpha$. Thus α is a root of $x^q - x$. So either g divides $x^q - x$ or by bezout's identity, $f(x)g(x) + g(x)(x^{p^n} - x) = \gcd(g, x^{p^n} - x) = 1$, (since g is irreducible) for some polynomials $f(x), g(x) \in F_p[X]$. In the second case, setting $x = \alpha$ we get $0 = 1$ which is a contradiction. Thus g is a factor of $x^q - x$. □

Problem 10. Consider the action of $G := \text{Gal}(F_q/F_p)$ on F_q . Explain why the orbits are in natural bijection with the irreducible polynomials over F_p whose degree divides r , where $q = p^r$ - with the orbit itself being the roots. How many irreducible polys of degree 5 are there over F_2 ? Do this without finding the polys.

Proof. Let S be the set of irreducible polynomials which divide r . Let $G = \text{Gal}(F_q/F_p)$. Let $\varphi : G/F_q \rightarrow S$, where $G * a \rightarrow \prod_{\alpha \in H * a} (x - \alpha)$. Let $\psi : S \rightarrow G/F_q$ where $f \rightarrow G * a$ where a is a root of f .

Suppose a is a root of f , $f(g(a)) = g(f(a)) \implies g(a)$ is also a root of $f \forall g \in G$. Since $\deg f \leq |H| = r$, all roots of f are in the orbit $H * a$.

$\forall f \in S$, $\varphi(\psi(f)) = \varphi(H * a) = f$. $\forall H * a \in G/F_q$, $\psi(\varphi(H * a)) = \psi(\prod_{\alpha \in H * a} (x - \alpha)) = Ha$

Thus S and G/F_q are in bijection.

All irreducible degree 5 polynomials are factors of $x^{2^5} - x$. The irreducible factors of $x^{2^5} - x$ have either degree 5 or 1. 0 and 1 are roots of $x^{2^5} - x$ so we can divide 2 linear terms and now all other factors of $x^{2^5} - x$ are all the degree 5 irreducibles. After dividing x and $x + 1$ we get a polynomial of degree 30. Thus there are 6 degree 5 polynomials. □